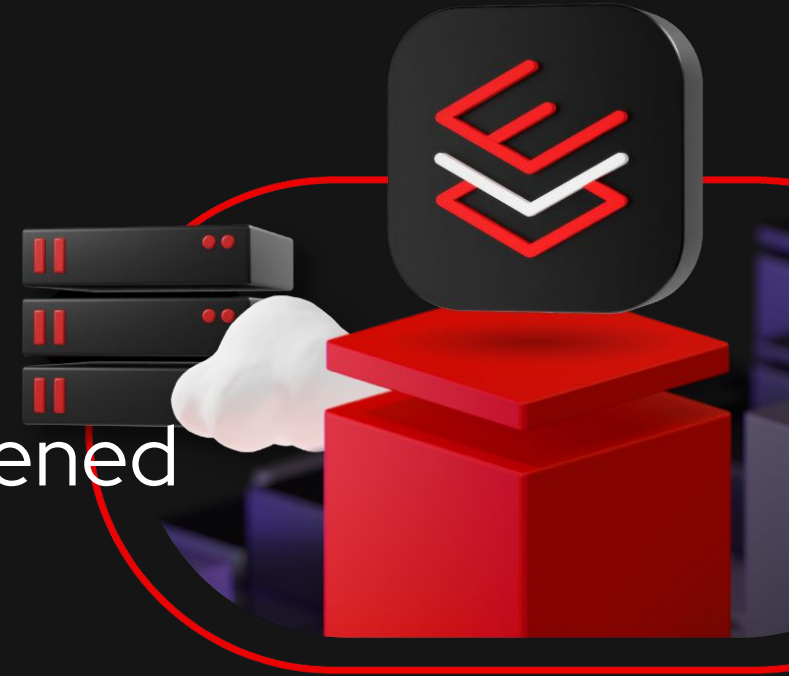


LB 2865

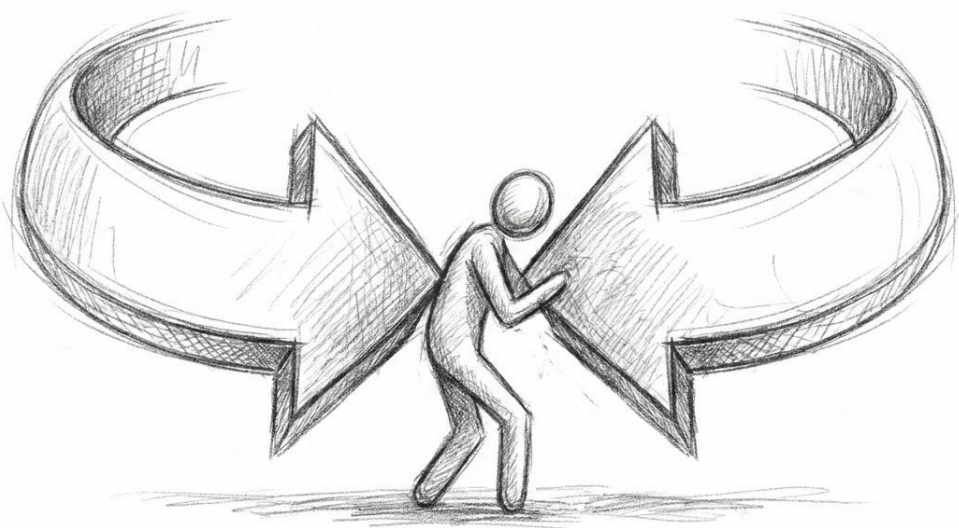
Container hardening

From zero to hero with Red Hat Hardened Images



The Security Squeeze

"The goal of cybersecurity is not to eliminate all risks, but to intelligently manage risks to an acceptable level." – *Michael Coates, former CISO of Twitter*



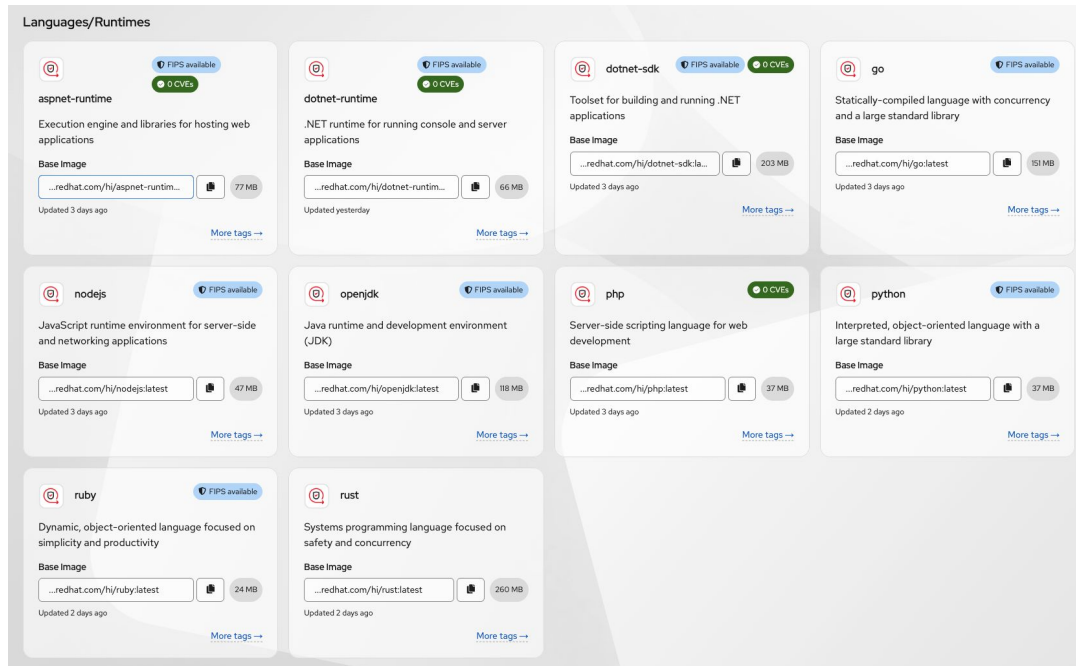
- **CVE Volume:** AI-driven discovery and automated scanning pushed CVE counts past ~60k annually¹. Traditional triage processes struggle to keep pace and discern the noise.
- **Compliance Pressure:** Regulatory pressures and risk profiles push towards to near-zero tolerance of vulnerabilities of any kind.
- **Innovation Tax:** Engineering teams are burning cycles on vulnerability maintenance and reporting instead of building features.





Red Hat Hardened Images

Modern software, delivered as hardened, verifiable container images



Foundational set of open source languages, runtimes, databases, web, & utilities aiming to reduce toil around CVEs.

- Built on a minimal, hardened Linux userspace optimized for containerized environments.
- Available at no cost. Support is included with Red Hat Enterprise Linux & OpenShift Container Platform
- Delivered through Red Hat's SLISA 3 trusted supply chain.



Guiding Principles

Foundation for Secure Cloud-Native Applications

Minimalism

Images contain only the absolute bare essentials and follow distroless principles to reduce the attack surface.

Hardened by Default

Packages are compiled with a full suite of modern security flags, with NIST 800-53 baseline controls applied where applicable.

Familiar Developer Experience

Images are designed for seamless integration with existing CI/CD pipelines and developer workflows.

Upstream Proximity

Security fixes are delivered directly from the latest upstream releases, targeting a time to remediate as soon as the fix is available, which typically is less than 24 hours for critical vulnerabilities.

Verifiable

Every image is delivered with a comprehensive Software Bill of Materials (SBOM) for transparency.



Hardened Images

at every level



Benchmarks & scanning
CIS, STIG, OpenSCAP profiles

Images
Distroless, non-root, reproducible

Packages - Hardened compiler options
PIE/RELRO, stack canaries, dep minimization

Source - Ever-fresh remediations
SLSA 3 provenance, continuous upstream tracking, 7-day crit SLA



The core of Hardened Images is Project Hummingbird

Let's take a closer look



Hummingbird RPMs

Fast. Minimal. Hardened packaging



A container-native packages - always up-to-date

- A rolling release model focused on the latest stable upstream
- Not a general-purpose distribution.
- Slimmed dependencies - drop features that are irrelevant for containers
- Stats:

Package Count: 85% reduction

Image Size: 60% smaller

CVE Count: Near-zero target

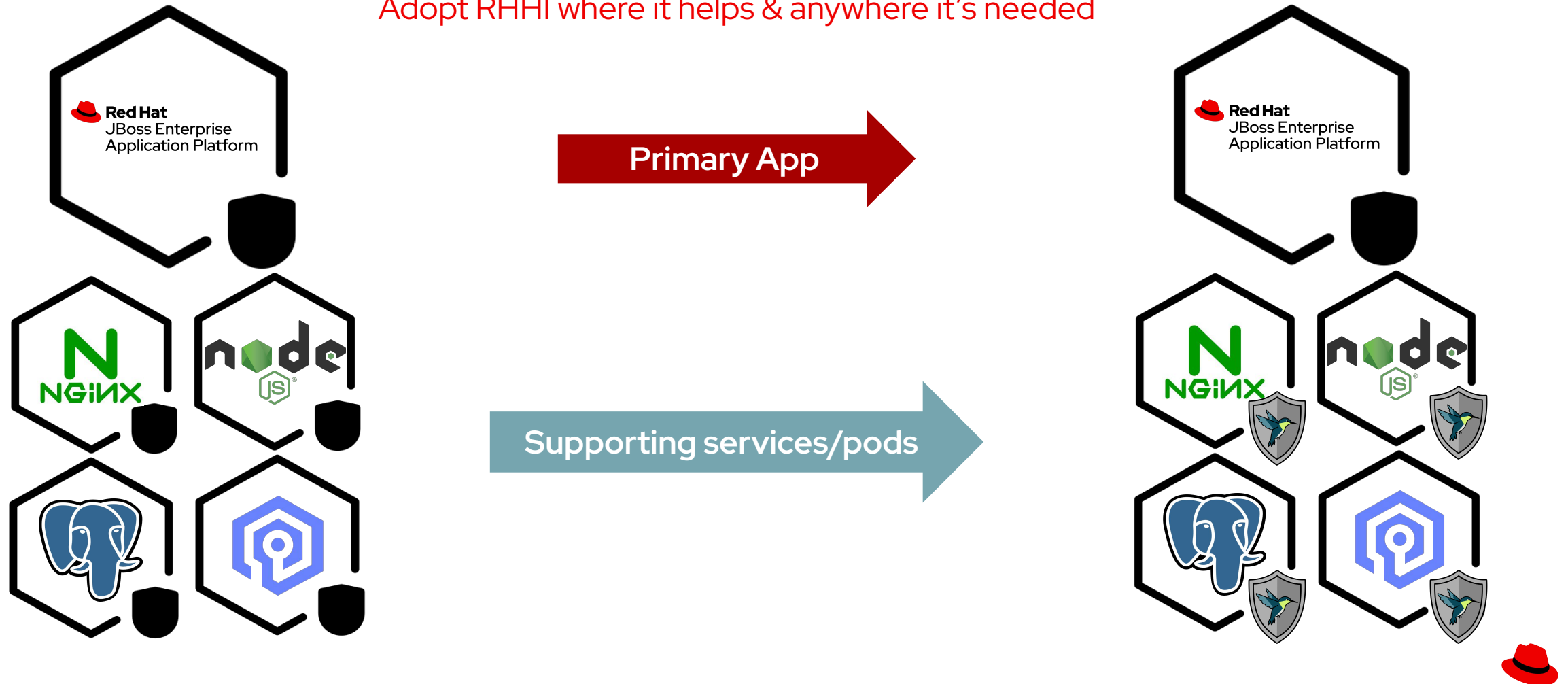
Update Frequency: <24hrs typical

Build Time: Hermetic, reproducible



Lock-down supporting pods

Adopt RHHI where it helps & anywhere it's needed



What you'll explore in today's lab

New web presence

Available images

Usage practices

Security workflows

Update patterns

Build automation

Secured pipelines



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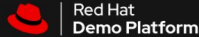
LB2865

Red Hat Hardened Images

May 11, 1:00 PM

[Access this lab →](#)





LB1750 - Advanced Features of Ansible Automation Controller

Access to LB1750 - Advanced Features of Ansible Automation Controller

Email * ⓘ

Workshop Password * ⓘ

Access this workshop →

The email you registered with

**Make sure you remember the (unique) e-mail address you use!
It will allow you to come back to your lab if you lose connection.**



Today's password is:

The screenshot shows a web browser displaying the Red Hat Hardened Images website. The browser's address bar shows the URL: `showroom-user-gxiz8-showroom.apps.ocp.f5bnh.sandbox5228.opentlc.com/?t=Red+Hat+Hardened+Images&p=index&w=33`. The website has a dark header with the Red Hat logo and the text "LB2865 Container Hardening: Zero to Hero". The main content area is titled "Red Hat Hardened Images" and features a large green arrow pointing to the "Request an image" button. Below this, there is a section for "Popular images" which lists several container images with their base images and sizes. The images listed are:

- go**: Statically-compiled language with concurrency and a large standard library. Base Image: `...redhat.com/hi/go:latest`. 151 MB. Updated 5 hours ago.
- mariadb**: Relational database management system compatible with MySQL. Base Image: `...redhat.com/hi/mariadb:latest`. 50 MB. Updated 5 hours ago.
- python**: Interpreted, object-oriented language with a large standard library. Base Image: `...redhat.com/hi/python:latest`. 38 MB. Updated 5 hours ago.
- aspnet-runtime**: Execution engine and libraries for hosting web applications. Base Image: `...redhat.com/hi/aspnet-runtime:latest`. 66 MB.
- dotnet-runtime**: .NET runtime for running console and server applications. Base Image: `...redhat.com/hi/dotnet-runtime:latest`. 66 MB.
- dotnet-sdk**: Toolset for building and running .NET applications. Base Image: `...redhat.com/hi/dotnet-sdk:latest`. 203 MB.
- go**: Statically-compiled language with concurrency and a large standard library. Base Image: `...redhat.com/hi/go:latest`. 151 MB.

The website also includes a "Contents" section on the left side, a "Request an image" button, and a "Product documentation" link. A large green arrow points to the "Request an image" button.

Or red.ht/hi in your browser



Where's the Info?



Finding secure images shouldn't be a scavenger hunt. If you have to "hunt" for the right bits in a crowd of noise, you're already losing time.



Scan & Spot Challenge

INSTRUCTIONS

Locate these 4 items at images.redhat.com

- >_ Pull Command
- ⚙ Exact CVE Count
- $\frac{1}{2}$ Variant Count
- 📄 SBOM & Verify Proof

TARGET IMAGE

MariaDB
latest

What is the number of CVEs + Variants?



FIRST 5 HANDS UP WITH CORRECT ANSWER WILL GET A HUMMINGBIRD GIFT





Red Hat Hardened Images

Our catalog of trusted, distroless, and micro-sized components m... vulnerability toil. Scanned for security and rigorously tested for fu... these hardened container images are available to you at no-cost.

Request an image

Product documentation →

Popular images



go

FIPS available

Statically-compiled language with concurrency and a large standard library

Base Image

...redhat.com/hi/go:latest



139 MB

Updated yesterday

More tags →



mariadb

:latest

FIPS only

Open in a new tab



Overview

Tags

SBOM

Verify Build

CVE Reporting

mariadb

MariaDB is a high-performance, relational database management system. Built on the MySQL technology, it offers a powerful SQL interface for data access and includes advanced features, such as support for multiple storage engines.

Stream

10.11

11.8

Variant

builder

default

Usage

Defining the MariaDB Root User Password

To use this image, you must set one of the following environment variables when you start the container:

- MARIADB_ROOT_PASSWORD=<password>: Sets the root password to the one you specified in this environment variable.
- MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=1: Sets an empty root password.
- MARIADB_RANDOM_ROOT_PASSWORD=1: Generates a random password for the root user and displays it in the container's log.

Basic Usage

For example, to start a MariaDB container with a defined root password and map the container port 3306 to a host port, enter:

```
podman run --detach --name <container name> --env
```

Start using this image

...redhat.com/hi/mariadb:latest



Updated yesterday



Use Builder version

Enables shell, package manager

54 MB



Architecture

amd64

arm64



1 CVE detected



grype 0.112.0 · Last scanned May 7, 2026 at 04:44 AM



Red Hat build of mariadb hardened image

Application category: Database & Data Management

Lightweight, security-focused container image maintained by Red Hat for production workloads.



Scan & Spot Challenge

INSTRUCTIONS

Locate these 4 items at images.redhat.com

- >_ Pull Command
- ⚙ Exact CVE Count
- 1/2 Variant Count
- 📄 SBOM & Verify Proof

TARGET IMAGE

core-runtime
FIPS 2.42

What is the number of CVEs + Variants?



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Request an image

Product documentation →

Popular images



go

FIPS available

Statically-compiled language with concurrency and a large standard library

Base Image

...redhat.com/hi/go:latest



139 MB

Updated yesterday

More tags →

Find by name

All types

Only FIPS available



core-runtime



FIPS only

Open in a new tab



Overview

Tags

SBOM

Verify Build

CV Reporting

core-runtime

The core-runtime image provides a base for running dynamically-linked applications. It includes essential libraries to execute programs that require a standard C library, such as those written in Rust, without extra development tools.

Stream latest

Variant

builder

default

openssl

openssl-builder

2 more

Usage

Use the default and -fips variants to run applications. For multi-stage builds, use the -builder variant to prepare artifacts for your final image. If your environment requires the openssl utility and libraries, select the -openssl variant.

Adding an Application by Using a Multi-stage Build

For production deployments, use a multi-stage build to create minimal container images. Compile your application by using an image that contains the required tools and use the core-runtime image to run the application:

```
# Build stage: Compile the Go application by using a Go container
FROM registry.access.redhat.com/hi/go:latest AS builder
ADD example.go .
RUN go build -o /tmp/example ./example.go
```

```
# Runtime stage: Copy the application to the runtime
FROM registry.access.redhat.com/hi/core-runtime:latest
COPY --from=builder /tmp/example /app
```

Start using this image

...redhat.com/hi/core-runtime:l...



Updated yesterday



Use Builder version

Enables shell, package manager

14 MB



Architecture

amd64

arm64



0 CVEs detected



grype 0.112.0 · Last scanned May 7, 2026 at 04:48 AM



Red Hat build of core-runtime hardened image

Application category: Operating System

Lightweight, security-focused container image maintained by Red Hat for production workloads.

Scan & Spot Challenge

INSTRUCTIONS

Locate these 4 items at images.redhat.com

- >_ Pull Command
- ⚙ Exact CVE Count
- 1/2 Stream Count
- 📄 SBOM & Verify Proof

TARGET IMAGE

OpenJDK 21
Runtime

What is the number of CVEs + Streams?



FIRST 5 HANDS UP WITH CORRECT ANSWER WILL GET A HUMMINGBIRD GIFT





Red Hat Hardened Images

Our catalog of trusted, distroless, and micro-sized components m... vulnerability toil. Scanned for security and rigorously tested for fu... these hardened container images are available to you at no-cost.

[Request an image](#)[Product documentation →](#)

Popular images



go

FIPS available

Statically-compiled language with concurrency and a large standard library

Base Image

...redhat.com/hi/go:latest

151 MB

Updated yesterday

[More tags →](#)

All types ▾

☐ Only FIPS available

openjdk

:21.0-runtime ▾



FIPS only

[Open in a new tab](#)

Overview

Tags

SBOM

Verify Build

CV Reporting

openjdk

OpenJDK provides the official reference implementation for Java SE to develop and run applications. It includes the JDK for building and the headless JRE for running Java applications.

Stream

21

25

Variant

builder

default

fips

fips-builder

4 more

Usage

Running Java Applications

The -runtime variant includes the runtime for running Java applications:

```
podman run --rm --volume <path_to_source_code>:/app:Z registry.access.redhat.com/hi/openjdk:latest-runtime java -jar /app/<application>.jar
```

Compiling Java Applications

The default variant includes the Java Development Kit and development tools, such as javac, jshell, and javadoc.

To compile a Java application, enter:

Start using this image

...redhat.com/hi/openjdk:21.0-r...

Updated 20 hours ago



Use Builder version

Enables shell, package manager

86 MB



Architecture

amd64

arm64



CVES detected



grype 0.112.0 · Last scanned May 7, 2026 at 04:45 AM



Red Hat build of openjdk hardened image

Application category: Programming Languages & Runtimes

Lightweight, security-focused container image maintained by Red Hat for production workloads.

Reducing Verification Toil



Zero-Auth Pulls

Podman/Docker pull commands
available without login gates.
Immediate access to bits.



Shift-Left Visibility

See the health index and CVE count
before pulling. No more manual
scan cycles.



Provenance Proof

Signed SBOMs ready for immediate
piping into your internal security
scanners.

"Security and provenance shouldn't be a scavenger hunt. It should be Ready-to-Use."



So where's the Info?





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It will allow you to come back to your lab if you lose connection.**



FileEditViewHistoryBookmarksToolsHelp

Virtual machines - mmi...Workstreams | Personal xbcvk-ephemeral | Man f xrhel-bootc-examples/se xrhel-labs/demo-kiosk: P xLB2865: Container Hard...

showroom-user-gxlz8-showroom.apps.ocp.f5bnh.sandbox5228.opentlc.com/?t=Terminal&p=index&w=33

mmicene.remote.csb | syncthing | rhel-asimov | Forgejo | Focalboard | Workstreams | Perso... | Receipts | TMM tools | BU tools | HR Tools | Image mode | migration | HUM

Red Hat SummitLB2865 Container Hardening: Zero to HeroTerminalRed Hat OpenShift ConsoleRed Hat Quay RegistryRed Hat Advanced Cluster SecurityRed Hat Hardened Images

Terminal

By continuing, you agree to use this environment solely for the steps in the lab guide. Any other use of this service and appropriate action will be taken with these terms you must disconnect now.

[lab-user: ~]\$

1. In the terminal window on your right log into OpenShift:

oc login -u user-gxlz8 -p i0hcy1acg0vZ https://api.ocp.f5bnh.sandbox5228.opentlc.com

2. Log into your OpenShift Console tab using the following credentials:

Username:

user-gxlz8

Password:

i0hcy1acg0vZ

The OpenShift console is also available in a separate window if you prefer

<https://console-openshift-console.apps.ocp.f5bnh.sandbox5228.opentlc.com>.

Click **Skip tour** if the windows appears.

3. For pushing images to registries **Red Hat Quay** is available in a tab on your right or if you prefer a separate window at <https://quay.apps.ocp.f5bnh.sandbox5228.opentlc.com>.

Username:

user-gxlz8

Password:

i0hcy1acg0vZ

OpenShift Console (Module 2)

InstructionsSplitTabs

26

The Red Hat logo, which is a red fedora hat with a black band and a gold buckle.

The Red Hat Hardened Images team wants to hear from *you!*



Help shape what comes next.

Scan the QR code or go to:

red.ht/userfeedback

Share your pain points, experiences, and ideas.

Visit the UX team at the RHEL booth

Sharpen your skills beyond the lab

Learn virtually and practice anywhere.

Gain the technical skills and knowledge you need to stay ahead of a growing, complex IT landscape through access to our comprehensive learning curriculum.



Give Red Hat Hardened
Images a try today



Visit red.ht/hi to explore





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